

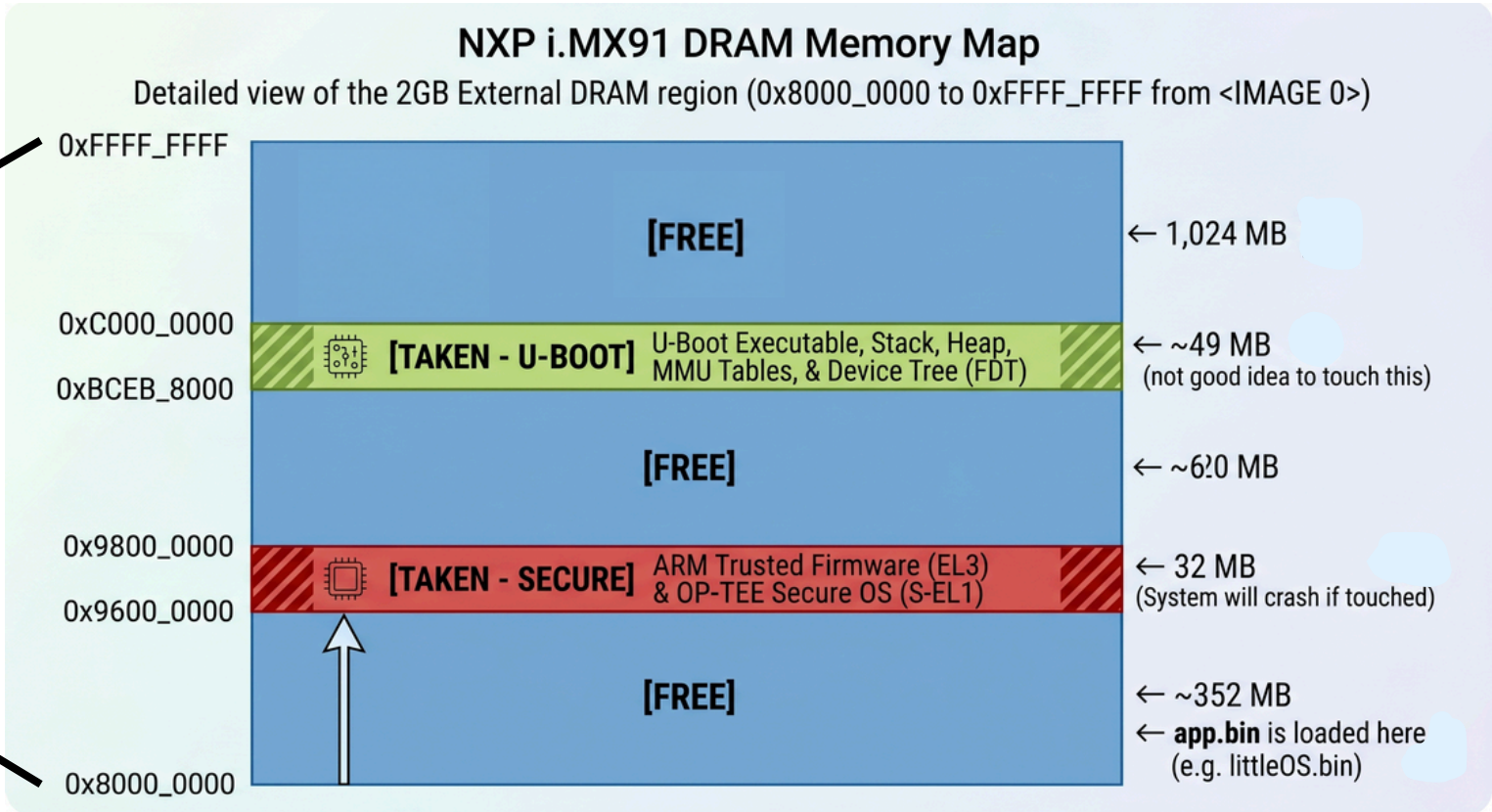
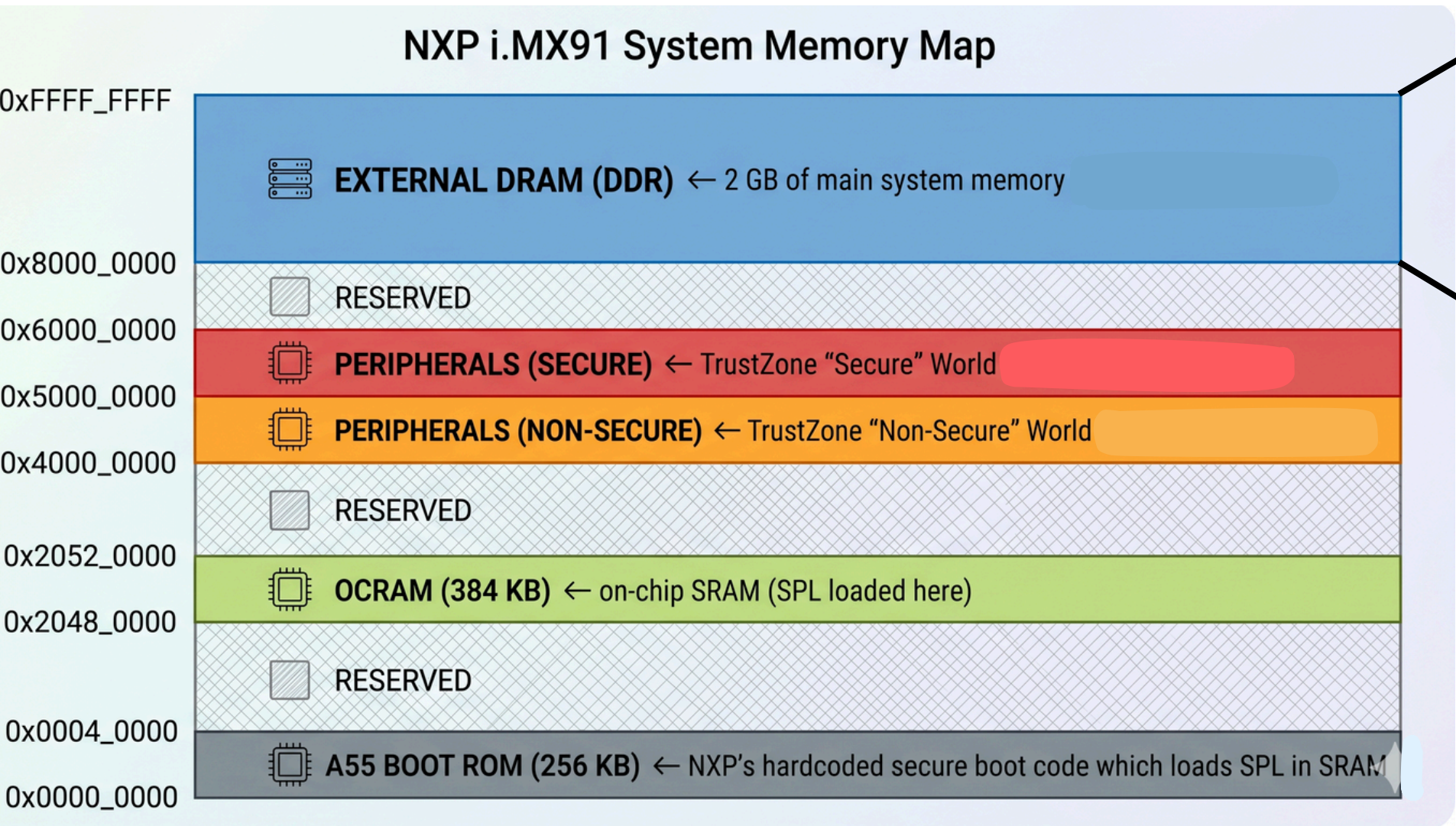
Exception Levels:

- lvl : what is accessible : while running what thing (example)
- ELO(NS) : can access only NON-SECURE memory : Standard User Apps (Web browser, etc)
- ELO(S) : can access both SECURE and NON-SECURE memory : Trusted Applications (Fingerprint matching, DRM)
- EL1(NS): can access only NON-SECURE memory : Rich OS Kernels (Linux, Windows, littleOS)
- EL1(S) : can access both SECURE and NON-SECURE memory : Trusted OS (OP-TEE, Trusty)
- EL2(NS): can access only NON-SECURE memory : Hypervisors (KVM, Xen) & U-Boot (pre-handoff)
- EL3(S) : can access both SECURE and NON-SECURE memory : ARM Trusted Firmware (TF-A) / Secure Monitor

Note:

- SRAM : on chip fast but little RAM
- DRAM : DDR RAM (2GB available here)
- BOOT ROM : Reads SW1 switches then load few bytes (SPL code) from selected BOOT Media into SRAM
- SPL (Secondary Program Loader) : Initialize DRAM, load U-Boot code in DRAM and jump to U-Boot code
- U-Boot : Loaded in DRAM (near 0xFD00_0000). Initialize clocks and all things for OS to boot

System Memory Map For NXP i.MX91



Images Generated using Gemini
but with actual memory locations
verified with manuals